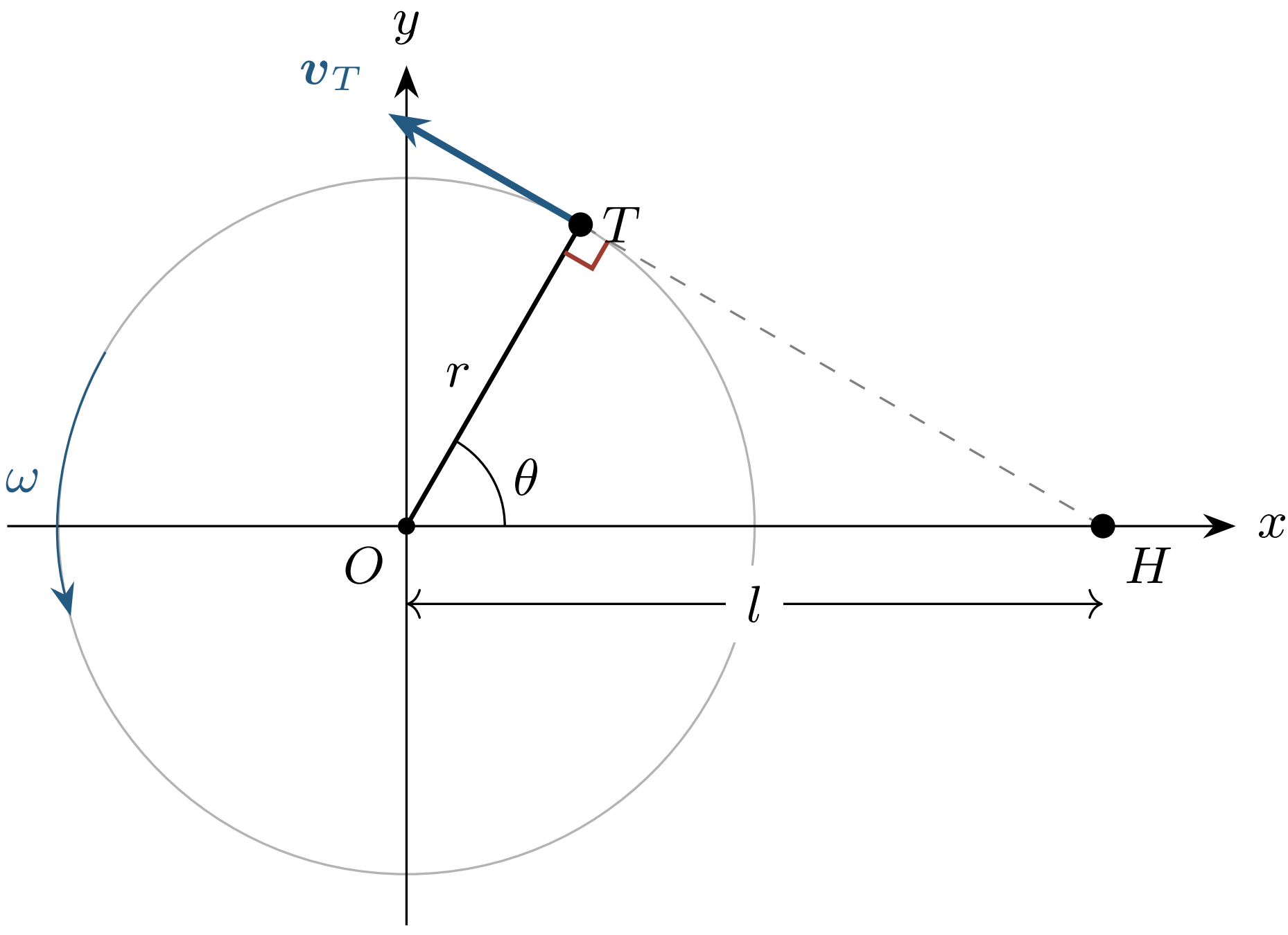
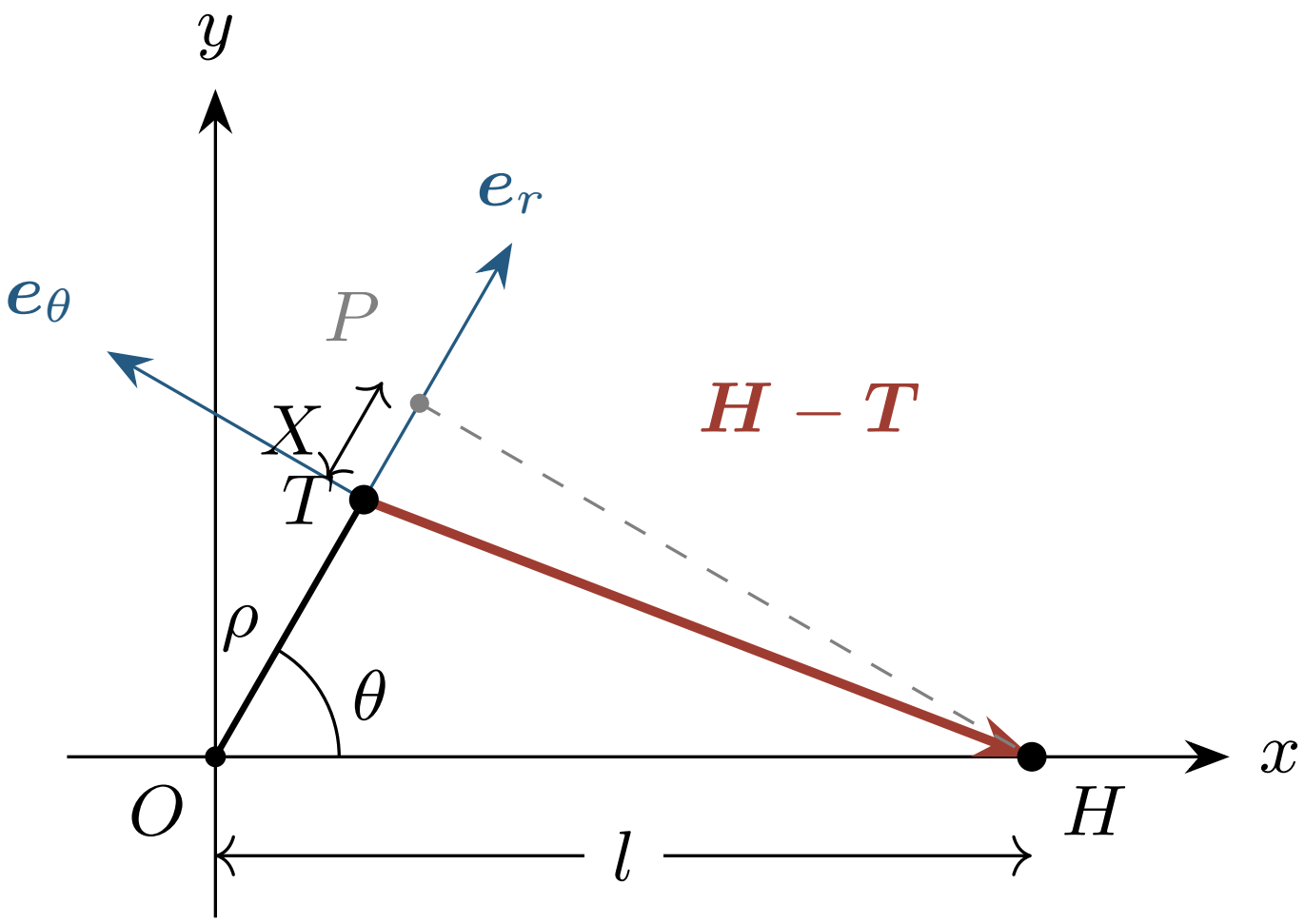
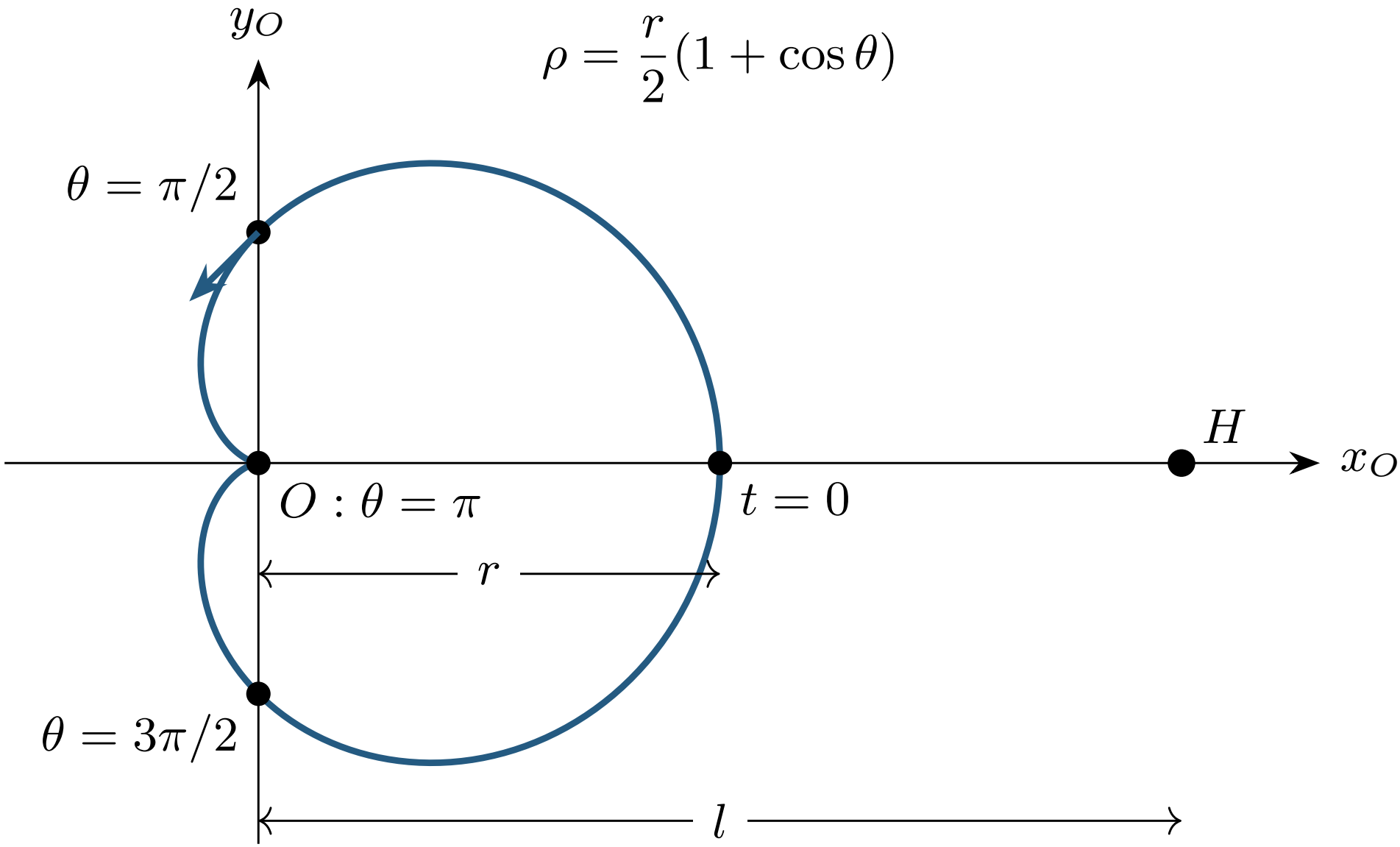


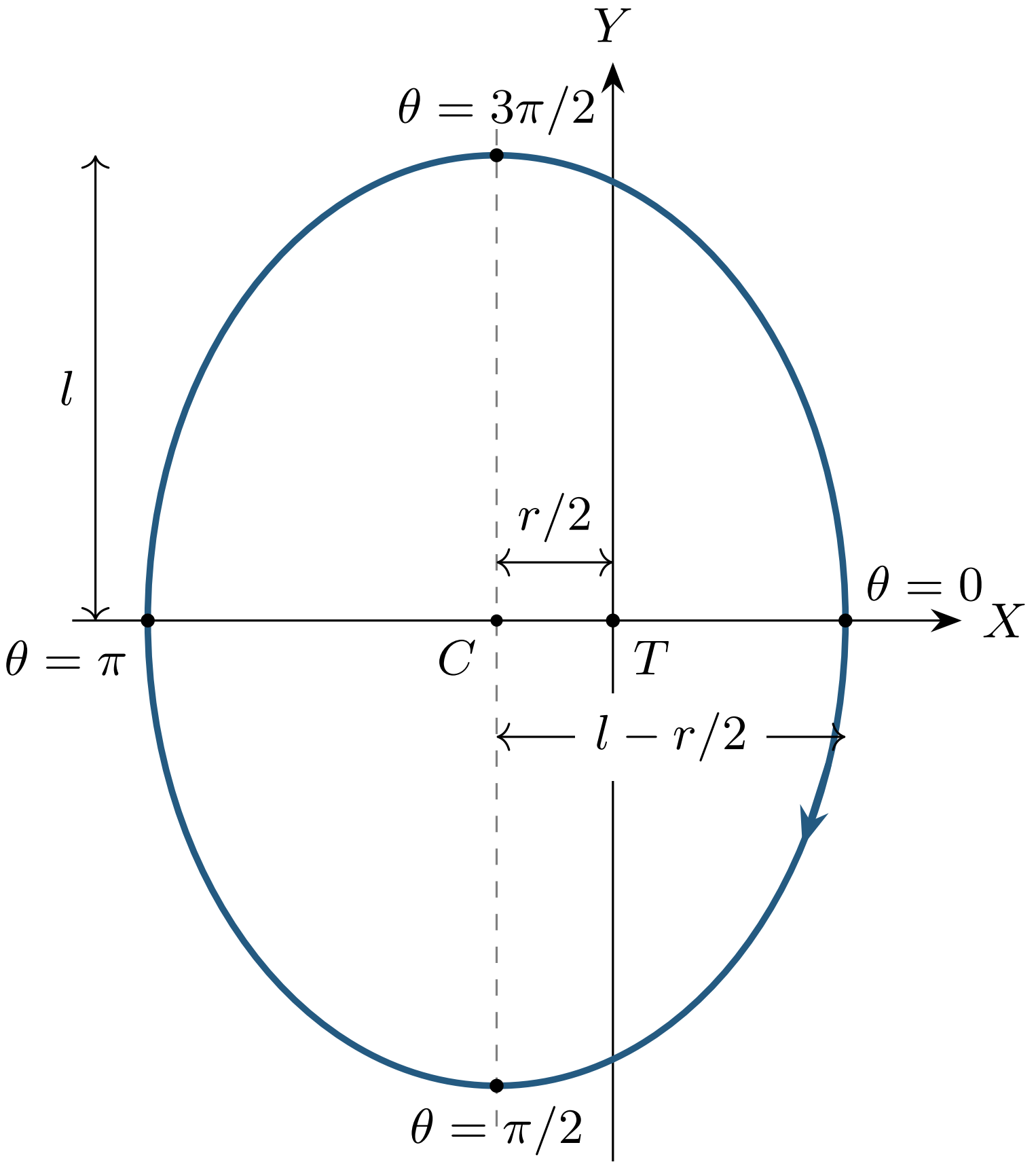
$$\boldsymbol{v}_T \parallel HT, \quad OT \perp HT$$



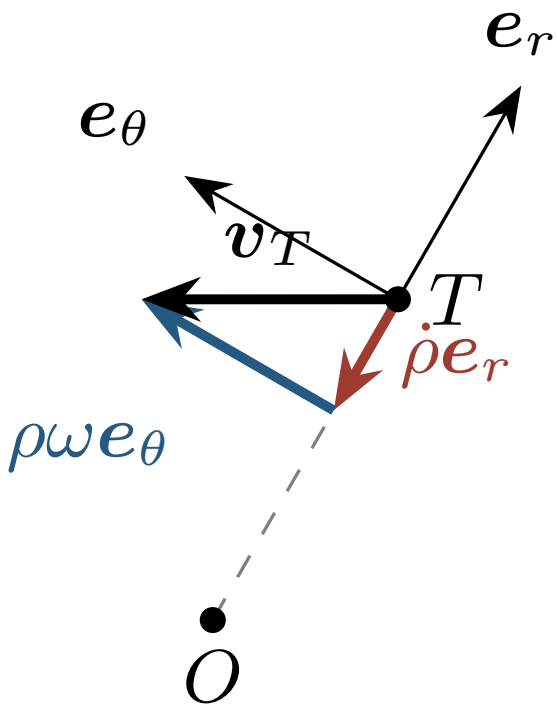
$$X = (\boldsymbol{H} - \boldsymbol{T}) \cdot \boldsymbol{e}_r$$

$$Y = (\boldsymbol{H} - \boldsymbol{T}) \cdot \boldsymbol{e}_\theta$$

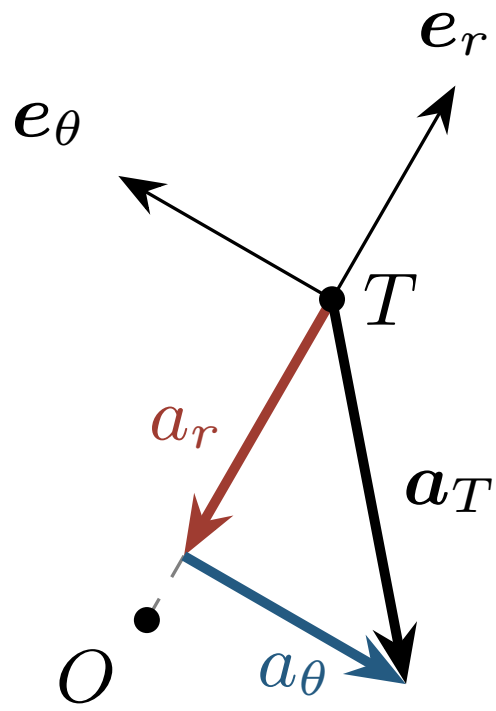

y_O
$\rho = \frac{r}{2}(1 + \cos\theta)$
$\theta = \pi/2$
H
x_O
$O : \theta = \pi$
$t = 0$
r
$\theta = 3\pi/2$
l



$C = (-r/2, 0)$，轨迹按顺时针方向遍历

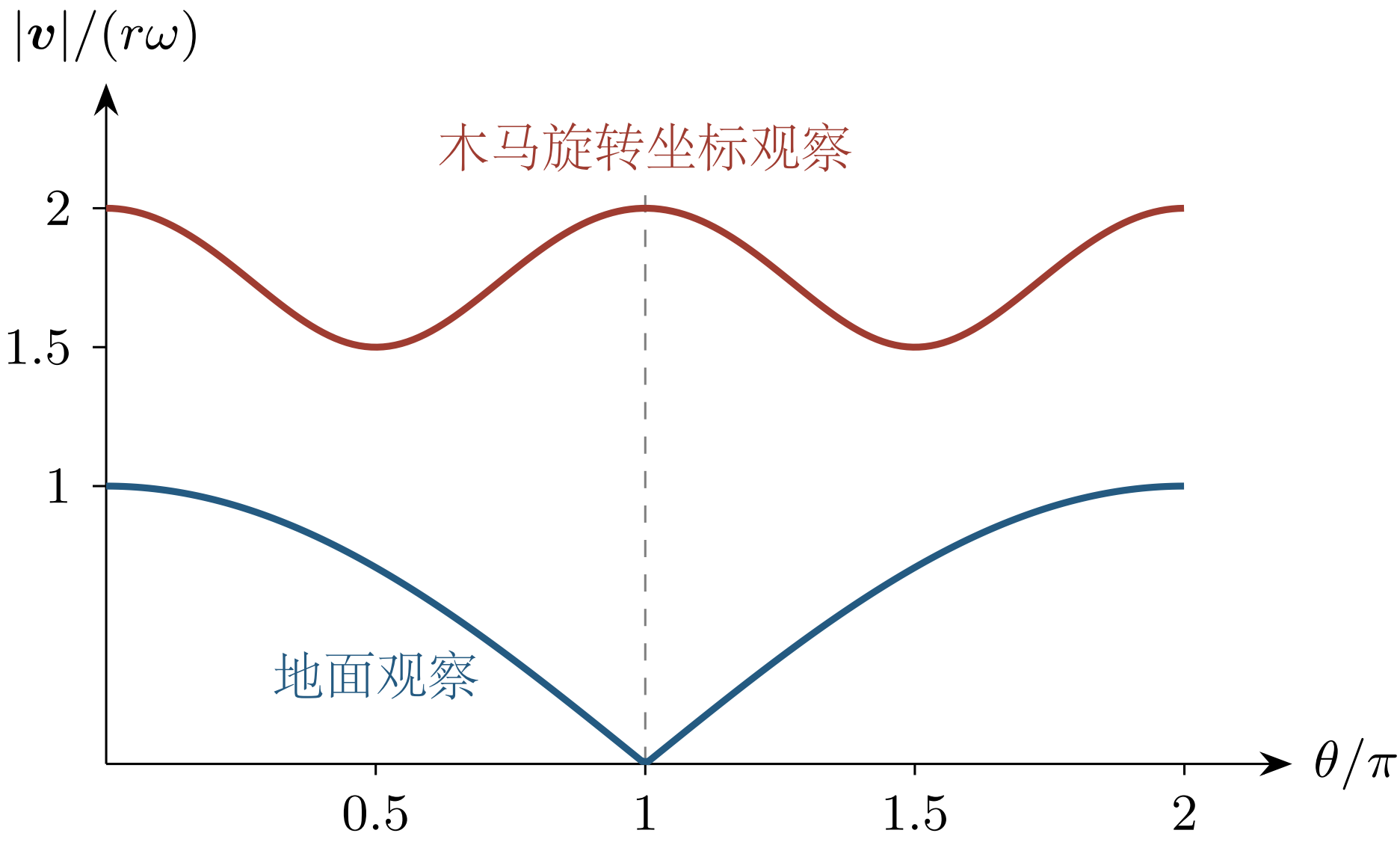


速度的径向与切向分解



$$a_r = \ddot{\rho} - \rho\omega^2$$

$$a_\theta = 2\omega\dot{\rho}$$



示例：$l = 2r$，$\theta = \omega t$